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Light-Activated Hybrid Nanocomposite Film for Water and Oxygen Sensing

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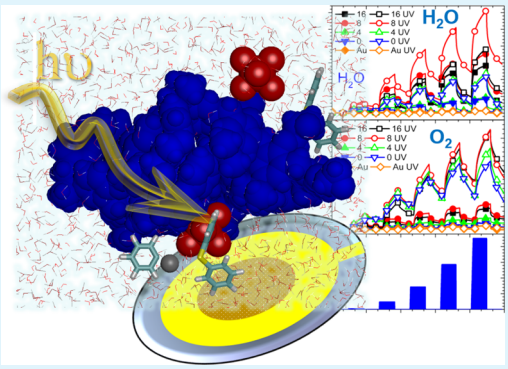
Supporting Information

ABSTRACT: Oxygen and water vapor sensing properties are investigated in metal-oxide−hybrid polymer nanocomposite thin films generated by infiltration synthesis, which incorporates molecular ZnO into the matrix of SU-8 polymer, a common negative-tone photoresist. The hybrid thin films display 20-fold higher gravimetric responses to oxygen and water vapor than those of control ZnO thin films in the dark. An additional 50−500% enhanced responses are detected under UV irradiation. The overall enhanced gravimetric response in the hybrid film is attributed to the ZnO molecules distributed in the polymer matrix, whereas the UV enhancement is explained by the light-induced, reversible generation of hydrophilic fluoroantimonic acid from triarylsulfonium hexafluoroantimonate photoacids, which leads to the increased surface potential and adsorption energies for oxygen and water. A gravimetric sensor based on a series of ZnO-infiltrated SU-8 films under UV excitation enables 96% accurate classification of water and oxygen environment with sub 10 mTorr detection limits. The results demonstrate UV-induced fully reversible surface hydrophilicity of ZnO/SU-8 hybrid nanocomposites.



KEYWORDS: *ZnO, SU-8, hybrid nanocomposite, QCM, atomic layer deposition, infiltration synthesis*

1. INTRODUCTION

The electrical and optical properties of ZnO, a wide and direct bandgap (∼3.37 eV) semiconductor, lead to the development of sensors for water,[1,2] oxygen,[3] volatile organic compounds (VOCs),[4−6] NO_x,[7] ammonia,[6] and UV light.[2]

Incorporation of ZnO nanoparticles in photolithographically patternable epoxy derivative of a bis-phenol-A Novolac (SU-8) leads to additional control of dielectric, elastic, or piezoelectric properties of polymer, while keeping composite compatible with complementary metal−oxide−semiconductor technology. ZnO−SU-8 composites showed Young modulus a factor of 4 higher than pure polymer[8,9] improving properties of low-cost piezoelectric microelectromechanical systems (MEMS) for applications in bio- and chemical sensing. SU-8 surface is not highly hydrophilic (static contact angle with water is about 80°), and for bio-MEMs applications the SU-8 usually UV-photo-grafted with 2-hydroxyethylmethacrylate or of poly-(ethylene glycol) hydrogels[10] to make it hydrophilic. Exposure to oxygen plasma etching can also increase surface roughness and improve hydrophilicity of SU-8 (reducing contact angle to 30°).[11]

Unlike SU-8, formation of highly hydrophilic (water contact angle smaller than 10°) surface of ZnO can be achieved under UV irradiation. This photowetting of ZnO is reversible, and it was successfully incorporated in a design of functional surfaces.[2,12,13]

Photoswitchable wetting was also reported for micro−nano-structure-layered system composed of TiO_2 nanorod-coated SU-8 micropillars, where TiO_2 enables photoswitching behavior, and SU-8 serves as passive macrostructural element.[14] No evidence that UV irradiation could affect contact angle of polymer without TiO_2 was reported.

We speculate that residual of photoacid generator (PAG), such as triarylsulfonium hexafluoroantimonate ($(Ar)_3S^+SbF_6^-$) used cross-linking SU-8 monomers and left in polymer after cross-linking, could lead to photowetting behavior, and thus enhanced sensing. UV light exposure could decompose the PAG to $H^+SbF_6^-$, which acts as a catalyst for cationic polymerization by opening SU-8 epoxy rings during thermal annealing, resulting in a chemically stable cross-linked SU-8 matrix.[15−17] $H^+SbF_6^-$ is strongly hydrophilic and forms fluoroantimonic acid hexahydrate ($H^+SbF_6^- + 6H_2O$) upon exposure to water,[18] but its reactivity as a superacid prevents utilization of $H^+SbF_6^-$ for common H_2O-sensing applications.

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To incorporate different concentrations of ZnO nanoparticles in SU-8 polymer film, we used atomic layer deposition (ALD). The ALD offers a low temperature alternative to conventional ZnO synthesis techniques, which require high temperatures (340−500 °C)[6,19] incompatible with polymer films. ALD also allows to increase the concentration of ZnO distributed in the polymer film matrix by increasing the number of infiltration cycles, a condition difficult to achieve for the conventional ZnO nanoparticle/SU-8 nanocomposite prepared by mixing ZnO and polymer monomers, due to strong UV absorption and scattering properties of ZnO. Although ALD-synthesized ZnO−SU-8 composites already been reported, no reliable information is given about distribution of ZnO in nanometer-thin film, which is critical to explain mechanism of sensing, mechanical properties of ZnO−SU-8 composites. Here, we address this unsolved question presenting analysis of the distribution of ZnO in ZnO−SU-8 composite as a function of ZnO-infiltration cycles.

The gravimetric response of ZnO-infiltrated SU-8 thin films deposited on a quartz crystal microbalance (QCM) to relative humidity (RH) and oxygen in the dark and under UV irradiation was investigated. The QCM is a thickness-shear mode resonator, which is used for a broad range of environmental sensing applications due to its subnanogram sensitivity and compatibility with gas, liquid, and vacuum environments.[20−23] Measurement of the dynamic frequency response of QCM coated with nanometer-thin polymer or oxide films allows in situ monitoring of analyte sorption kinetics[24,25] and provides a platform for testing effects of UV activation on gravimetric response of adsorbent films.[20] Integration of basic ZnO oxide nanoparticles in SU-8 matrix is expected to increase surface area and either passivate superacid or enhancement interaction with water due to photowetting of ZnO.[26−29]

Hybrid nanocomposite prepared by ALD infiltration synthesis of molecular ZnO in SU-8 thin films exhibits a significantly enhanced gravimetric response to water vapor and oxygen compared with a control samples of pure SU-8 or ZnO thin film. Further enhancement of sensor response is realized by exposing films to UV irradiation. Through the complementary structural, physical, and chemical analyses, the enhanced gas sensing functionality is explained by mechanism of the increased number of adsorption sites originating from the infiltrated ZnO along with which leads to reversible formation of fluoroantimonic acid and subsequent enhancement of composite's affinity for oxygen and water molecules. The results also demonstrate a new sensing functionality realized by infiltration synthesis, but also reveal a sensing mechanism based on reversible UV-induced chemical changes, which may enable enhanced control of the gas/vapor sensor selectivity and sensitivity.

2. EXPERIMENTAL SECTION

2.1. Chemicals and Materials. Commercially available SU-8 2002 solutions (MicroChem) diluted in cyclopentanone (Sigma-Aldrich) were used for experiments. The diluted SU-8 solution was spin-cast at 2000 rpm for 45 s to prepare ∼100 nm thick SU-8 films on precleaned 5 MHz AT-cut Au-coated QCM crystals, SiO_2/Si substrates with prepatterned 2 μm spaced interdigitated electrodes, and bare Si substrates. The spin-cast films were subjected to the ambient pre-cross-linking baking for 1 min at 65 °C and additional 3 min at 95 °C on a hot plate (i.e., soft bake). The samples were then exposed to UV light for 5 min under nitrogen using a UV light chamber equipped with a low-pressure Hg lamp (American Ultraviolet Co.), followed by the postexposure bake for 2 min at 95 °C in ambient air to complete the cross-linking. The film thickness was measured by an interferometer (Filmetrics).

2.2. ZnO-Infiltration Synthesis. The ZnO-infiltration synthesis was performed on the spin-cast SU-8 thin films at 85 °C in a commercial ALD system (Cambridge Nanotech Savannah S100) using the procedure described in our previous report.[29] Diethylzinc, DEZ (Aldrich), and water were used as the organometallic Zn precursor and oxidant. Each ZnO-infiltration synthesis cycle consists of (a) exposure of SU-8 films to DEZ for 300 s under the static vacuum (chamber pressure ∼ 2 Torr); (b) purge of the reactor by 100 sccm N_2 flow for 60 s; (c) exposure of SU-8 films to water vapor for 300 s under the static vacuum (pressure ∼ 10 Torr); and (d) purge of the reactor by 100 sccm N_2 for 300 s. 4−16 cycles of ZnO infiltration to the samples. Control ALD-mode ZnO thin films were prepared by simply placing bare inorganic substrates (e.g., Au-coated QCM crystals) with no SU-8 film coating during the infiltration synthesis on "hard" inorganic substrates, the precursors cannot diffuse into the substrates, only allowing the typical, surface-limited normal ALD with a deposition rate of ∼1 Å/cycle.[28,29]

2.3. Cross-Sectional Transmission Electron Microscopy (TEM). SU-8 thin film was examined after 10 cycles of ZnO-infiltration synthesis by cross-sectional TEM (JEOL 1400, 120 kV). The cross-sectional TEM sample was prepared by the lift-out technique in a focused ion beam system (FEI Helios) using evaporated carbon and top Au protective layers applied on top of the sample.

2.4. Atomic Force Microscopy (AFM) and Kelvin Probe Force Microscopy (KPFM). All AFM and KPFM measurements were performed on a commercial AFM (Cypher, Asylum Research and Oxford instr. Company), using as-received Pt/Ir-coated AFM probes (Nanosensors, PPP-EFM), with a nominal mechanical resonance frequency and spring constant of 75 kHz and 2.8 N/m, respectively. Measurements were performed using amplitude modulated detection in lift mode, or two pass modes, with the lift height being between 50 and 100 nm ensuring the response was purely electrostatic in nature.

2.5. Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS). Time-of-flight secondary ion mass spectrometry (ToF-SIMS) measurements were performed in positive ion detection mode using a TOF-SIMS-NSC (ION-TOF GmbH) instrument inside an ultrahigh vacuum chamber at ∼6.4×10^{-6} Torr. Liquid metal Bi_3^+ ion gun (30 keV energy, 30 nA current) was used as a primary source with a focused ion beam spot size of ∼5 μm, and mass resolution $m/\Delta m$ ∼ 5000. Cs^+ ion beam with energy of 1 keV and current 70 nA was used for depth profiling with spot size ∼20 μm. Sputter crater size was 300 μm. Measurements were done in non-inter-laced mode, when every scan by the bismuth ion beam was followed by a 1 s sputtering cycle. Cs_2OH^+ and Cs_2F^+ clusters were used to detect spatial distribution of negative ions of OH^- and F^-, respectively.

2.6. Raman Spectroscopy. ZnO-infiltrated hybrid SU-8 films on SiO_2/Si substrates were characterized in 100−7500 cm^{-1} frequency range using inVia Qontor confocal Raman spectrometer (Renishaw equipped with 300 l/mm grating). Raman and fluorescence spectra of thin films were measured in backscattering geometry through a 20× objective using 633 nm laser operated at 5 mW. The final spectrum is an average of seven spectra taken at different locations on the sample.

2.7. Contact Angle Measurements. The contact angles of deionized water on the ZnO-infiltrated hybrid SU-8 films with varying numbers of infiltration synthesis cycles were measured by a KRÜSS DSA 30 contact angle goniometer in the dark and under UV irradiation produced by an Hg lamp (254 nm, see Figure S1 for the measured spectral irradiance) placed 5 cm away from the film surface.

2.8. Electrical and Gravimetric Measurements. All electrical and gravimetric measurements were performed at 27 ± 0.5 °C inside a modified controlled environmental chamber (Surface Measurement Systems Ltd). Prior to gas/vapor adsorption measurements, films were placed under 10^{-5} Torr (1.33×10^{-5} mbar) vacuum for 12 h. QCM frequency shift was recorded using a commercial QCM system (SRS QCM200). Gas/vapor pressure was controlled by injecting O_2/

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H_2O into the vacuum chamber using a mass flow controller at a 20 cm^3/min flow rate. A LabVIEW program was used to control gas/vapor flow and record frequency shift of the QCM during the experiment. An Hg pen lamp was mounted outside the vacuum chamber through a quartz window for UV irradiation experiments. The spectral irradiance of UV lamp (Figure S1) was measured by an optical power meter (Newport 2931-C with 918D UV detector head). Direct-current (DC) electrical resistance measurements were performed with a source-measure unit (Keithley 2420).

2.9. Density Functional Theory (DFT) Calculations. To evaluate the relative interaction strengths of water with fluoroantimonic acid and triarylsulfonium hexafluoroantimonate, DFT calculations were performed using the Vienna ab initio simulation package.[30] The Kohn–Sham equations were solved using the projector augmented wave approach and a plane-wave basis with a 400 eV energy cutoff. The exchange–correlation interactions were considered in the generalized gradient approximation using the Perdew–Burke–Ernzerhof functional.[31] To account for nonlocal correlations (dispersion interactions), the vdW-DF2 approach was used.[32] Electronic convergence was defined as a consistency between successive cycles of less than 10^{-5} eV. Supercells consisting of a water molecule and either fluoroantimonic acid or triarylsulfonium hexafluoroantimonate with a vacuum layer of at least 18 Å were used to avoid spurious interactions with periodic images. Full relaxation (geometry optimization) was performed with a *K*-point sampling restricted to the γ point, which is appropriate for such finite cluster calculations. All atoms were relaxed until the residual forces were below 0.01 eV/Å. The interaction energies were then determined by computing the difference between the total energy of the clusters and the individual molecular constituents.

3. RESULTS AND DISCUSSION

3.1. Microstructure and Chemical Depth Profile of ZnO-Infiltrated Hybrid SU-8 Films. The cross-sectional TEM examined the microstructure of 10 cycle ZnO-infiltrated SU-8 film, confirming the infiltration of molecular ZnO within the matrix SU-8 film. The high-contrast bright-field cross-sectional TEM image (Figure 1a, left panel) features a dark contrast in the 40–50 nm depth of the upper portion of SU-8 film, originating from the high atomic number of Zn in the ZnO-infiltrated polymer matrix. A higher magnification image (Figure 1a, right panel) further reveals that the infiltrated ZnO does not form any visible crystallinity or aggregation in the polymer matrix while resulting in a 4–5 nm thick dense layer atop the film. The observed infiltration depth, amorphous nature of the ZnO-infiltrated hybrid, and dense top layer are all consistent with our earlier study.[12] The cross-sectional TEM images indicate a seemingly smooth surface in the given field of view, and the AFM examination of the top surface (Figures S2, taken from eight cycle ZnO-infiltrated SU-8) reveals some sporadic small spheroidal domains of 5–20 nm radius on the film surface, but otherwise a background with a roughness comparable to that of the control ZnO film simultaneously formed on a separate substrate without the SU-8 coating during the infiltration synthesis (root-mean-square roughness < 1 nm).

We further investigated the chemical depth profiles of SU-8 film after 0, 4, 8, and 16 cycles of ZnO-infiltration synthesis using ToF-SIMS. Figure 1b shows the average chemical depth profiles for SU-8 films with 4, 8, and 16 ZnO-infiltration cycles with standard deviation represented by error bars. Sputter time corresponds to film depth, where the time value of 0 s corresponds to the hybrid film surface. The intensity of the Si^+ peak suggests that the SiO_2/Si substrate surface was reached after 150–200 s of sputtering. Peaks attributed to Cs_2F^+, Cs_2OH^+, and $C_2H_5^+$ were observed throughout the film. The

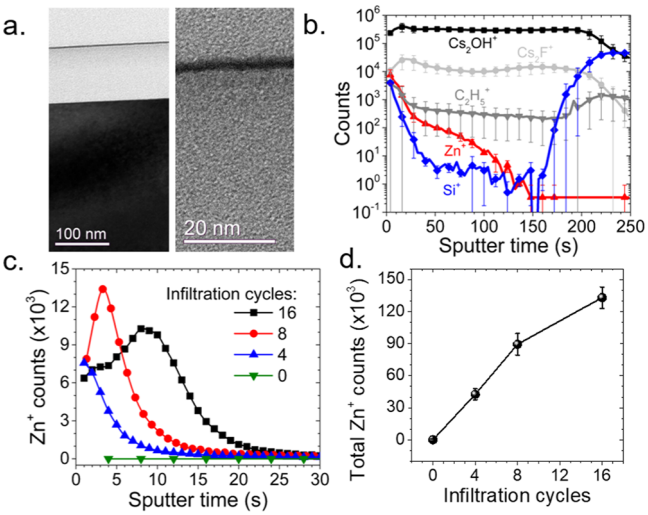

Figure 1. (a) Cross-sectional bright-field TEM images of SU-8 film after 10 cycles of ZnO-infiltration synthesis, with the left panel showing a low-magnification, high-contrast view of the whole hybrid film and the right panel the upper portion of the film including the top dense layer at a higher magnification. (b) Intensity of ToF-SIMS peaks attributed to Cs_2OH^+, Cs_2F^+, $C_2H_5^+$, Zn^+, and Si^+ measured as a function of sputter time. Error bars represent the standard deviation between four different films. Sputter time is correlated to film depth, where 0 s corresponds to the film surface. (c) Profile of ToF-SIMS peak attributed to Zn^+ in films with 0, 4, 8, and 16 infiltration cycles. (d) Total Zn^+ counts, which is directly proportional to concentration of Zn in films, with 0, 4, 8, and 16 infiltration cycles.

presence of fluoride can be attributed to the PAG, which is uniformly distributed through the film thickness. We also detected the presence of $C_2H_5^+$ ions with heightened intensity near the surface, most likely due to the terminal ethyl group of ZnO precursor, diethylzinc (DEZ), resulting from its incomplete reaction with water (i.e., C_2H_5–Zn–C_2H_5 + H_2O → C_2H_5–Zn-OH + C_2H_6), or the reaction byproduct C_2H_6.[28,29] The presence of Cs_2OH^+ throughout the whole film thickness indicates an abundance of hydroxyl terminations within the film, which can originate from both the cross-linking reaction of epoxy groups in SU-8 during the initial film preparation and the reaction between DEZ and water during the infiltration synthesis.[18] Given the largely constant depth profile of Cs_2OH, the former is the more likely cause of the presence of hydroxyl groups, which in fact can serve as adsorption sites for H_2O and O_2 adsorbates. We detected the highest concentration of Zn^+ near the film surface, corresponding to the 4–5 nm thick dense layer observed in the cross-sectional TEM. The dense layer is attributed to the repeating infiltration synthesis cycles that start blocking pores in the polymer matrix, limiting the continued diffusion of material precursors down into the bulk film. To examine the effect of the number of infiltration cycles on the depth profile of infiltrated ZnO in SU-8, Zn^+ distributions are shown in Figure 1c. SU-8 film with 0 ZnO-infiltration cycle served as a reference. As the number of infiltration cycles increased, the infiltrated ZnO appears distributed more from the surface to the interior of SU-8 thin film. Zn^+ concentration decreased sharply after ~20 s of sputtering, suggesting that the highly ZnO-rich region extended from the surface down to ~15% of the film thickness depth. The area under the Zn^+ peak was used to estimate the total abundance of infiltrated ZnO in the film (Figure 1d). Zn^+ peak area increased roughly linearly with

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the number of infiltration cycles. A consistent, linear increase of the amount of infiltrated ZnO within the SU-8 film with increasing number of infiltration cycles has been previously observed by in situ mass gain measurements, which identified the mass gain of 0.68 $\mu g/cm^2$ per infiltration cycle, equivalent to the ZnO thickness of ~10.2 Å per cycle (assuming the density of ZnO is 5.61 g/cm^3), a more than 10-fold larger deposition rate than that of the normal ALD process, highlighting the distinct deposition mode of infiltration synthesis.[29]

3.2. Gravimetric Gas Sensing Properties of ZnO-Infiltrated Hybrid SU-8 Films. To investigate the effect of ZnO infiltration on gas sensing properties of SU-8 films, the hybrid films prepared on QCM crystals were exposed to controlled pressure pulses of oxygen and water vapor at room temperature in the dark. Control ZnO films deposited by ALD mode on polished gold-coated QCM crystals without SU-8 templates exhibited low gravimetric response (~1 Hz/cm^2, corresponding to ~20 ng/cm^2 under Sauerbrey conditions[33]) during exposure to 22.6 Torr pulses of O_2 and H_2O vapor (Figure S3a). The response of ZnO films was similar to that of bare gold reference crystals, which indicates a lack of selective H_2O and O_2 adsorption on ZnO likely due to the low surface area and lack of reactive ZnO terminations (e.g., unhydroxylated ZnO(1010) lattice planes containing O or Zn vacancies)[2] resulting from the deposition of ZnO under the normal ALD mode. Frequency shift was roughly linear with gas/vapor pressure, for the partial pressure range from 0.24 Torr (~1% RH) to 22.6 Torr (~95% RH) (Figure S3b), consistent with findings of Zhang et al. who reported the linear gravimetric RH response from ZnO nanowire/nanorod-coated QCM crystals in the 5−97% RH range.[34] We note that although the UV activation was previously reported to enhance ZnO gas/vapor response,[35] the effects of UV were not investigated on the control ALD-deposited ZnO films due to the lack of selective H_2O and O_2 adsorptions.

In contrast to the control ZnO film, the ZnO-infiltrated hybrid SU-8 exhibited remarkably enhanced gravimetric sensing responses to H_2O and O_2, which was further increased under UV irradiation. The frequency shifts of ZnO-infiltrated SU-8 films on QCM in response to pulses of H_2O/O_2 in the dark and under UV irradiation are shown in Figure S4. ZnO-infiltrated SU-8 films featured up to a factor of 20 higher gravimetric response (~20 Hz/cm^2) to H_2O (top panel, Figure S4a) and O_2 (middle panel, Figure S4b) than the control ZnO films on bare QCM crystals (Figure S3). Considering that the insignificant difference in surface roughness between the ZnO-infiltrated SU-8 hybrid and control ZnO films, the order of magnitude higher gravimetric responses was ascribed to higher analyte accessibility to active ZnO sites within the bulk of ZnO-infiltrated hybrid film, along with the presence of additional adsorption sites associated with hydroxyl groups in the SU-8 matrix.

More detailed gravimetric response characteristics of ZnO-infiltrated SU-8 films are found in the normalized temporal frequency shift measured during exposure to 22.6 Torr pulses of H_2O (corresponding to ~95% RH) and O_2 under continuous UV irradiation and in the dark (Figure 2a,b). Frequency shifts and response times (corresponding to the exponential decay constant to be discussed later) generally increased by a factor of 2 or more during UV irradiation, particularly during O_2 exposure most likely due to the photoactivation of O_2 adsorption sites on ZnO.[2] Exposure to

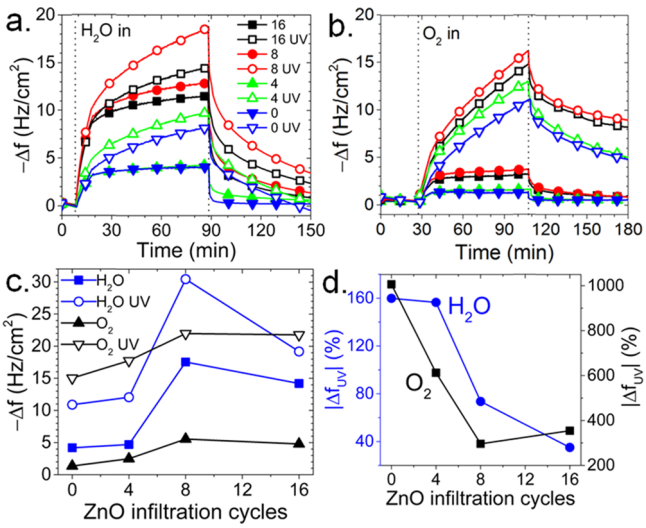

Figure 2. Normalized temporal frequency shift (Δf) of ZnO-infiltrated SU-8 films during exposure to (a) 22.6 Torr (~95% RH) pulse of H_2O vapor and (b) 22.6 Torr pulse of O_2. The data are normalized to Δf = 0 Hz at time = 0 s. Gas/vapor exposure and removal are marked by the vertical dotted lines. Numbers in legend correspond to number of ZnO-infiltration cycles and whether response was measured in the dark or under UV irradiation. (c) Frequency shift vs number of ZnO-infiltration cycles during exposure to H_2O and O_2 in the dark and under UV irradiation. Magnitudes shown in (c) are slightly higher than those in (a) and (b) because they were extracted from the absolute frequency shift (Figure S4). (d) Percent change in frequency shift ($|\Delta f_{UV}|$) between films exposed to gas/vapor in the dark and those measured under UV irradiation. Left and right axes show differences in frequency shift when films were UV irradiated during exposure to H_2O vapor and O_2, respectively.

UV irradiation increased gravimetric response of ZnO-infiltrated films to O_2 by a factor of 5 compared with its response to oxygen in dark. Density functional theory (DFT) calculations that included self-consistent van der Waals corrections also indicated a substantial increase in the interaction of a water molecule with fluoroantimonic acid, ~0.36 eV over triarylsulfonium hexafluoroantimonate. Additionally, a hydration shell of six water molecules was found to be easily supported by fluoroantimonic acid, corroborating the experimental results for the UV effect and the hypothesized mechanism. The infiltration-cycle-dependent frequency shift in response to 22.6 Torr gas/vapor pulses (Figure 2c) further shows that the gas/vapor response of the hybrid film increased with increasing number of ZnO-infiltration cycles from 0 to 8, likely resulting from an increase in the number of gas/vapor adsorption sites associated with molecular ZnO imbedded within the SU-8 matrix, such as hydroxyl groups and oxygen vacancies.[1,2,36,37] The SU-8 film with 16 ZnO-infiltration synthesis cycles showed smaller gas/vapor response than the film, which underwent 8 infiltration cycles, which suggests that repeated infiltration cycling and subsequent ZnO accumulation in the top dense layer of hybrid film eventually restricts analyte diffusion into the ZnO/polymer network. Uninfiltrated control SU-8 films exhibited only ~4 Hz/cm^2 frequency shift (~70 ng/cm^2 under Sauerbrey conditions) during exposure to water, comparable to the previously reported value for SU-8, but ~5-fold lower than that of the ZnO-infiltrated hybrid film, thus showing that the infiltrated ZnO within SU-8 matrix plays a critical role in enhancing the gravimetric response of ZnO-infiltrated SU-8 hybrid film to gas/vapor. It is noted that water

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vapor is known to exhibit Fickian diffusion and relatively low adsorption in pure SU-8 thin film.[38]

3.3. Origins of UV-Activated Gas Sensing in ZnO-Infiltrated Hybrid SU-8 Films. The effect of UV irradiation was further investigated and correlated with the presence of PAG in the SU-8 matrix and the UV-blocking properties of ZnO. To examine the effect of ZnO infiltration on UV enhancement of gravimetric response, the percent change in frequency shift due to UV irradiation ($|\Delta f_{UV}|$) is shown in Figure 2d. During both H_2O and O_2 exposures, the effect of UV light decreased with increasing number of ZnO-infiltration cycles, suggesting that the UV enhancement of gravimetric response is originating from the infiltrated film bulk as the highly ZnO-concentrated, top dense layer in the hybrid film can act as an effective UV-blocking layer protecting the underlying polymer.[39] The notion is further supported by the fact that the ZnO thin film itself has been shown to have a UV-enhanced H_2O/O_2 adsorption;[2,36] with increasing number of ZnO-infiltration cycles, the top dense layer on hybrid film is becoming more ZnO concentrated, but the UV enhancement was decreasing.

We suspect the most likely chemical entity in the bulk of hybrid film that can induce the observed UV-activated adsorptions of H_2O and O_2 is the decomposition/activation of residual PAG in the SU-8 matrix into $H^+SbF_6^-$, which can form fluoroantimonic acid hexahydrate ($H^+SbF_6^- + 6H_2O$) upon exposure to water.[18] Our previous study did show the presence of residual PAGs in the ZnO-infiltrated SU-8.[28,29] Considering that the light-induced activation of $H^+SbF_6^-$ should affect the surface energy, H_2O contact angle measurements were performed on the ZnO-infiltrated SU-8 hybrid film in the dark and under UV irradiation (Figures 3a and S5). The

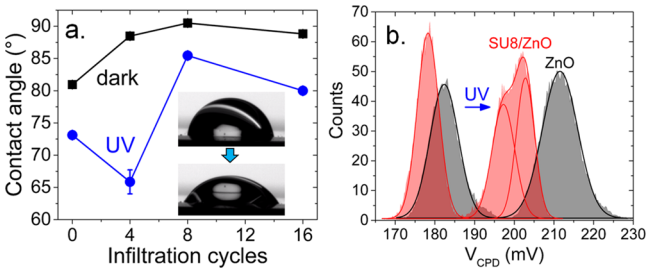

Figure 3. (a) Contact angle of water droplet measured on SU-8 films with 0, 4, 8, and 16 ZnO-infiltration cycles in the dark and under UV irradiation. Inset shows representative images of droplets in the dark (top) and under UV irradiation (bottom). Contact angles were measured after 2 min of droplet formation. (b) Histograms of contact potential difference (V_{CPD}) measured using KPFM in the dark and under UV irradiation for eight cycle ZnO (black) and eight cycle ZnO-infiltrated SU-8 (red).

contact angle in the ALD-mode-deposited ZnO reference film decreased by ~19° upon exposure to UV irradiation (not shown), whereas that in the ZnO-infiltrated SU-8 film decreased by an average of ~10° upon exposure to UV irradiation. We infer that the UV response in uninfiltrated film is likely related to the presence of PAG in the SU-8 matrix, which reversibly decomposes to hydrophilic $H^+SbF_6^-$, increasing the affinity of ZnO-infiltrated SU-8 hybrid films to water and oxygen. The onium salt PAG in SU-8, $(Ar)_3S^+SbF_6^-$, generally decomposes to acidic $H^+SbF_6^-$ by UV irradiation, catalyzing the SU-8 polymerization under thermal baking.[17] Fluorine detected throughout the whole thickness of ZnO-infiltrated SU-8 hybrid film by ToF-SIMS (Figure 1b) confirms that PAGs are present in abundance at the surface and interior of the SU-8 film after ZnO-infiltration synthesis. We note that the UV-induced superhydrophobic to superhydrophilic transition in ZnO thin films has been attributed to the photodesorption of oxygen previously, which results in hydrophilic hydroxyl terminations.[37] This can be one factor underlying the UV-enhanced hydrophilicity observed in ZnO-infiltrated films, but does not explain the increased hydrophilicity in uninfiltrated control SU-8 films.

The UV-induced changes in the surface energies of control ZnO and ZnO-infiltrated SU-8 hybrid films in the dark and under UV irradiation were further probed using KPFM (Figures 3b and S6). UV irradiation resulted in a ~16% increase in contact potential difference (V_{CPD}) in ZnO film and a ~12% increase in surface potential of ZnO-infiltrated hybrid SU-8 film. The value observed from the control ZnO film is in agreement with UV-induced V_{CPD} increase reported in ZnO nanowires via the photodesorption of oxygen from the ZnO surface.[3] V_{CPD} in ZnO consists of roughly Gaussian distribution of potentials, which undergoes a ~30 mV energy shift upon UV irradiation (black curves in Figure 3b). In the ZnO-infiltrated SU-8 film, the UV irradiation results in splitting of the single Gaussian V_{CPD} distribution into at least two distinct levels, which are shifted by ~19 and ~25 mV (red curves in Figure 3b). One of the peaks likely corresponds to energies associated with UV-activated ZnO. The other peak is not present in the ZnO film and must be associated with the SU-8 matrix, but only appears during UV irradiation. This suggests that it is related to ionic byproducts of $(Ar)_3S^+SbF_6^-$, which reversibly decomposes each time the film is exposed to UV irradiation.

3.4. Adsorption Characteristics of H_2O and O_2 in ZnO-Infiltrated Hybrid SU-8 Films. To further study gas/vapor adsorption on ZnO-infiltrated SU-8 hybrid films, the magnitudes of frequency shifts after 80 min of exposure to H_2O and O_2 pulses are plotted against gas/vapor pressure (Figure 4a). The gravimetric responses to gas/vapor pressure closely resemble modified Toth isotherms, shown fitted to experimental data by the solid lines ($r^2 > 0.98$ for each fit, where r is the residual in data fitting). Data were fitted by nonlinear least squares optimization using the SciPy library in Python.[40] Other common isotherms, including Langmuir, Temkin, and Freundlich models, were fitted and resulted in less suitable fits (average $r^2 \leq 0.9$).

The Toth isotherm is an empirical Langmuir-type model modified for heterogeneous adsorption on rough surfaces that yields an asymmetrical distribution of adsorption energies.[41] The Toth model is used for describing a broad range of interactions because of its predictive behavior from low (Henry-type limit) to high (saturation limit) pressures[42] and its suitability for systems limited to submonolayer coverage.[43] A modified Toth isotherm can be represented by

$$M = \frac{K_T p_R}{[1 + (K_T p_R)^t]^{1/t}} \quad (1)$$

where fitting parameters K_T and t are the Toth isotherm constant and Toth exponent, respectively. Total adsorbate mass (M) and relative analyte pressure (p_R) were normalized prior to fitting to produce unitless K_T and t parameters. Total adsorbate mass was calculated by converting frequency shift to mass using the Sauerbrey equation and normalized by M =

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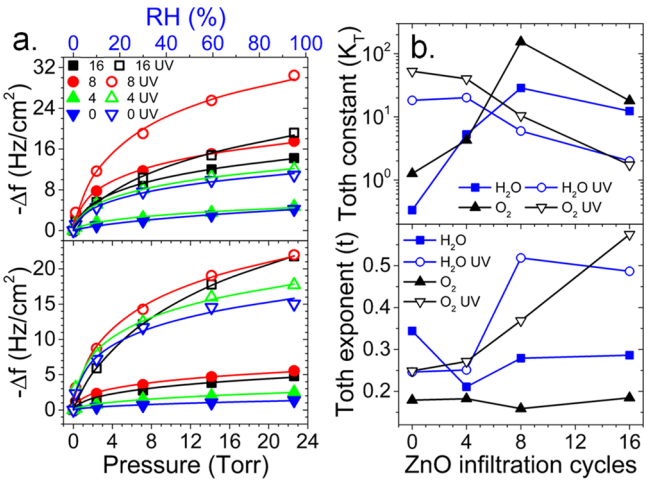

Figure 4. (a) Frequency shifts of ZnO-infiltrated SU-8 films after 80 min of H_2O (top panel) and O_2 exposures (bottom panel) at different gas/vapor pressures and illumination conditions (dark or UV). RH corresponding to each H_2O vapor pressure is shown on top horizontal axis. The solid lines correspond to fits to the Toth isotherm. Error bars correspond to noise signal in frequency shift measurements. (b) Corresponding Toth isotherm constants (K_T, top panel) and Toth exponents (t, bottom panel) extracted from Toth isotherm fits in (a). Note the use of logarithmic scale in top panel. Error bars correspond to uncertainties propagated from noise signal in frequency shift measurements.

$\Delta M/M_0$, where ΔM is the measured mass change shown in Figure 4a, and M_0 is the maximum mass change, which occurred in each film at the highest analyte pressure (22.6 Torr). Relative analyte pressure was defined as $p_R = p/p_0$, where p is the analyte partial pressure and p_0 is the partial pressure of H_2O at saturation at 27 °C ($p_0 \approx 23.8$ Torr). The Toth isotherm constant K_T corresponds to the adsorption affinity of a specific adsorbent−adsorbate pair and is related to its average adsorption energy. The Toth exponent t corresponds to homogeneity of the adsorbent surface, where $0 < t \leq 1$. For example, a completely homogeneous surface has $t = 1$, and the corresponding Toth isotherm reduces to the well-known Langmuir equation.[44]

To investigate effects of ZnO-infiltration synthesis on the adsorption affinity and heterogeneity of SU-8 films, the values of K_T and t were extracted from the Toth isotherm fits. Uninfiltrated SU-8 films exhibited a high adsorption affinity (K_T) under UV irradiation, likely due to photoactivation of hydrophilic $H^+SbF_6^-$ as discussed (Figure 4b, top panel). K_T decreased by roughly an order of magnitude as the number of ZnO-infiltration cycles increased, which likely occurs as UV is being blocked by the dense layer of ZnO accumulated near the film surface. In the dark, K_T was enhanced by increasing the number of ZnO-infiltration cycles (up to eight) due to the increase in available active adsorption sites associated with infiltrated ZnO. For the film with 16 ZnO-infiltration cycles, the ZnO accumulation near the surface likely blocks pores and prevents diffusion of analytes into the bulk film, thus reducing the average K_T.

The Toth exponent t, corresponding to film homogeneity, did not show a strong dependence on the number of ZnO-infiltration synthesis cycles when adsorption occurred in the dark (Figure 4b, bottom panel). This is expected since the majority of gas/vapor adsorption was associated with the photoactivated $H^+SbF_6^-$. In the dark, the value of t is generally larger for H_2O adsorption than for O_2 adsorption, indicating greater heterogeneity in O_2 adsorption sites than those of H_2O. This suggests that H_2O adsorption in the dark is primarily mediated by physical dipole and van der Waals interactions, for example between H_2O and hydroxyl groups or previously adsorbed H_2O molecules.[1] Meanwhile, O_2 adsorption may be associated with a range of different adsorption energies including those corresponding to the dissociative chemical adsorption at oxygen vacancies on the ZnO surface. Under UV irradiation, t increases by a factor of ∼2 as the infiltration synthesis increased from 0 to 16 cycles. Homogenization of the adsorption surface in the hybrid matrix is expected to occur as the SU-8 matrix is infused with the infiltrating molecular ZnO, and the hydrophilic fluoroantimonic becomes inaccessible to adsorbate molecules. The homogenization of adsorption sites on ZnO can also occur because the photodesorption of oxygen is known to result in hydrophilic hydroxyl terminations on ZnO surface.[37] It is noted that the Toth isotherm adsorption behavior observed in ZnO-infiltrated SU-8 films stands in contrast to the linear frequency shift observed in control ALD-mode ZnO films prepared on bare Au-coated QCM crystals (Figure S3b) as well as the linear response observed by Zhang et al. from the ZnO nanowire/nanorod-coated QCM crystals in the 5−97% RH range.[34] This provides further evidence that the nonlinear gas/vapor gravimetric response of ZnO-infiltrated SU-8 hybrid films (Figure 4a) can be attributed primarily to the presence of adsorption sites associated with SU-8.

3.5. Performance of Gravimetric Gas/Vapor Sensing of ZnO-Infiltrated Hybrid SU-8 Films. To assess the performance of ZnO-infiltrated hybrid SU-8 films as adsorbent materials for gas/vapor sensors, detection limits were estimated from the signal-to-noise ratio and magnitude of QCM response (Figure 5). Minimum detection limits for both O_2 and H_2O vapor generally decreased by an order of magnitude or more when the ZnO-infiltration synthesis was

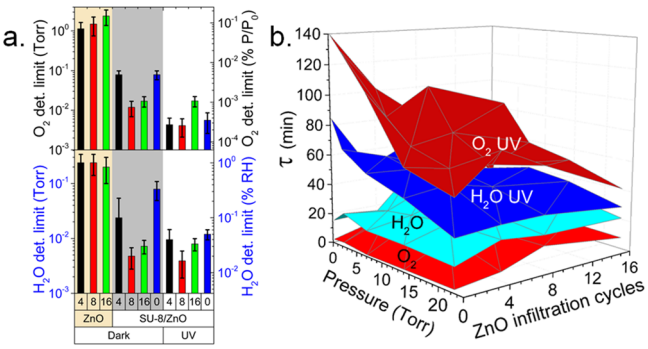

Figure 5. (a) Estimated detection limits for O_2 (top panel) and H_2O vapor (bottom panel) sensing using ZnO-infiltrated hybrid SU-8 films. The left vertical axes correspond to O_2/H_2O partial pressures, and the right vertical axes O_2 partial pressure in units of % P/P_0 (where P_0 is partial pressure of oxygen in atmosphere, ∼160 Torr) (top) and % RH at 27 °C (bottom). Note the use of log scales. The yellow shaded region corresponds to the control ALD ZnO films, the gray region to ZnO-infiltrated SU-8 films in the dark, and the white to ZnO-infiltrated SU-8 under UV irradiation. The numbers on horizontal axis correspond to the number of ZnO-infiltration synthesis cycles. Error bars correspond to amplitude of noise in frequency shift measurements. (b) Time constant (τ) measured from frequency shift during the adsorption of H_2O/O_2 in the dark and under UV irradiation.

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applied to SU-8 thin films, highlighting the enhanced sensing performance enabled by more available adsorption sites in the hybrid film. The limits further decreased by a factor of ∼2 under UV irradiation, driven by the activation of hydrophilic fluoroantimonic acid and photodesorption of oxygen from ZnO-adsorption sites, as discussed previously.[2] Under UV irradiation, SU-8 film with eight ZnO-infiltration synthesis cycles achieved the lowest detection limits corresponding to ∼0.004 Torr O_2 and ∼0.02% RH and showed potential for wide-range humidity sensing from 1 to 95% RH. The ZnO-infiltrated hybrid SU-8 film enabled significantly enhanced RH sensitivity (bottom panel) under a broad humidity range compared with the reported performances of other metal oxides (sensitive to 2% RH in 5−99% RH range),[45] graphene oxides (sensitive to 4% RH under 11−95% RH range),[46] organic small molecules (sensitivity of 2.5% in 0−95% RH range),[47] and conjugated polymers (sensitive to 1% RH in 33−97% range).[48,49]

In the dark, gravimetric responses of ZnO/SU-8 nanocomposite films were completely reversible after 1 h of gas/vapor desorption. Under UV irradiation, responses of ZnO/SU-8 films were over 80% reversible after 1 h of water desorption, and ∼50% reversible after 80 min of oxygen desorption (Figure 2a,b). The increased analyte adsorption under UV irradiation requires a longer desorption period, resulting in the lack of full reversibility between gas/vapor pulses under UV light. Improving the reversibility of ZnO-based gas sensors is generally achieved by thermally induced desorption, which may be investigated for the hybrid nanocomposite films in future studies.

To examine the kinetics of gas/vapor adsorption, the frequency shift of ZnO-infiltrated SU-8 films during O_2/H_2O exposure was fit to an exponential model

$$\Delta f = \Delta f_0 + A\, e^{-t/\tau} \tag{2}$$

where τ is the characteristic time constant for adsorption and A and Δf_0 are scaling and offset constants. The gravimetric response during gas/vapor adsorption is well explained by the exponential model ($r^2 > 0.985$) at each pressure under each environmental condition. The value of τ extracted during adsorption in the dark and under UV irradiation is shown in Figure 5b. Under UV irradiation, τ generally decreased with gas/vapor pressure, which suggests that the adsorption rate is diffusion-limited.[50] The average value of τ, after ∼60 min under UV irradiation and ∼20 min in the dark, is roughly an order of magnitude greater than the response times reported for other ZnO nanostructure-based sensors, including those for humidity and VOC, which typically exhibit response and recovery times on the order of 2 min[6,34,51] and the response time of capacitive SU-8 humidity sensors increasing with increasing level of cross-linking in the SU-8 matrix.[17] The large τ values measured here likely result from low analyte diffusion rates through the film due to a combination of the ZnO accumulation and the high level of polymerization in the SU-8 matrix. Under UV irradiation, τ generally decreased for higher numbers of ZnO-infiltration synthesis cycles likely due to more available infiltrated ZnO.

The clear separation between τ surfaces in Figure 5b, which correspond to each of the four tested environmental conditions, suggests that τ can be used for distinguishing between different ambient environmental conditions based on the response of each film. Classification of the four environments was carried out using a support vector machine (SVM) algorithm with radial basis function kernel. The SVM was implemented using the Scikit-learn module in Python using 5-fold cross-validation to prevent overfitting.[52] Using the number of ZnO-infiltration cycles, the gas/vapor pressure, and the value of τ as input variables, the SVM classified each of the four environments (O_2 in the dark, O_2 under UV irradiation, H_2O in the dark, and H_2O under UV irradiation) with over 96% accuracy. Considering the relatively small number of training points for the SVM (i.e., 5 gas/vapor pressures, 4 different infiltration cycles, 20 values of τ for each environment), it is expected that the environmental classification can reach 100% accuracy with a larger quantity of training data. The results demonstrate the possibility of using the ZnO-infiltrated hybrid SU-8 films with different levels of infiltration as sensing arrays for accurate classification of ambient environmental conditions.

3.6. Optical and Electrical Properties of ZnO-Infiltrated Hybrid SU-8 Films. Finally, the effects of ZnO-infiltration synthesis on the optical and electrical properties of SU-8 films were investigated. Along with the demonstrated enhancement in gravimetric gas/vapor sensing performance, the information should guide the potential application of ZnO-infiltrated hybrid SU-8 films as multifunctional sensing media. The confocal Raman spectroscopy of ZnO-infiltrated SU-8 films (on SiO_2/Si substrates; 633 nm excitation wavelength) featured a broad photoluminescence band at 1690 and 2450 cm^{-1} (∼4.1−5.9 μm wavelength, Figure S7a), which has been associated with defect states in bulk and epitaxial ZnO[53] or quantum-confined ZnO nanoparticles.[54] To further confirm the effect of ZnO-infiltration synthesis on the photoluminescence spectra of SU-8 films, the mean-centered principal component analysis (PCA) was performed on the spectra shown in Figure S7a. A nonlinear iterative partial least squares algorithm with leverage correction validation was used for decomposition of eigenvectors during PCA. Two principal components (PC1 and PC2) explained 98% of the differences between experimental spectra (Figure S7b). Films without the low-frequency (1690−2450 cm^{-1}) feature in the Raman-photoluminescence spectra are clustered near the intercept corresponding to PC2 = 0 (red region), whereas the SU-8 films with ZnO infiltration contain the contributions from both components (blue region) in the spectral features. There is a clear distinction between ZnO-containing SU-8 films, pure ZnO, SU-8, and SiO_2, which suggests that photoluminescence from ZnO-infiltrated SU-8 is influenced by morphological or optoelectronic interaction between ZnO and SU-8. The higher PC2 contribution to spectral features of 8 cycle ZnO-infiltrated SU-8 compared to that of 16 cycle-infiltrated SU-8 suggests that interaction between infiltrated ZnO and the SU-8 matrix is not strongly correlated with the number of ZnO-infiltration synthesis cycles. Multivariate curve resolution (MCR) performed without an initial guess on photoluminescence spectra revealed two optimal spectral components (Figure S8), in agreement with results of PCA. The MCR enabled separation of two spectral features in the photoluminescence spectra, which are likely associated with ZnO species (low-frequency feature with a peak around 1450 cm^{-1}).

We investigated the electrical properties of ZnO-infiltrated hybrid SU-8 films by measuring the DC electrical resistance in ambient air in the dark and under UV irradiation. In the dark, the DC resistance of SU-8 film is ∼10^8 Ω (Figure 6a). After four ZnO-infiltration synthesis cycles, the resistance decreased

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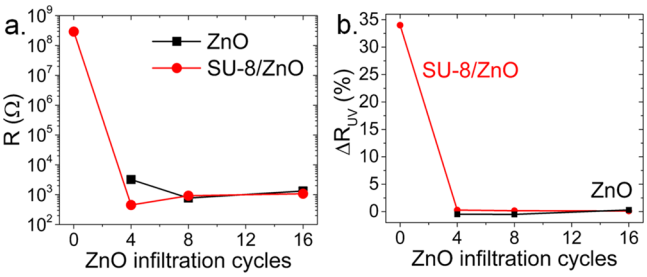

Figure 6. (a) DC electrical resistance of control ALD-mode-prepared ZnO films and ZnO-infiltrated hybrid SU-8 films measured in the air in the dark as a function of the number of infiltration synthesis cycles. The ZnO-infiltrated SU-8 film with 0 infiltration cycle corresponds to the uninfiltrated SU-8 film. (b) DC electrical resistance change (ΔR_{UV}) after 10 min of UV irradiation for control ZnO films and ZnO-infiltrated hybrid SU-8 films.

by ∼5 orders of magnitude to ∼10^3 Ω. Subsequent infiltration cycles did not significantly alter the resistance, which suggests that four cycles of infiltration resulted in the formation of conductive ZnO pathways, allowing electrical conduction. After 8 and 16 infiltration cycles, the resistance of ZnO-infiltrated film was roughly equal to that of the control ALD-mode-prepared ZnO films, likely because of the highly ZnO-concentrated top dense layer formed at the hybrid film surface. Interestingly, there was a UV-induced increase in the resistance of ZnO-infiltrated SU-8 film, whose extent was diminishing for the larger number of infiltration synthesis cycles (Figure 6b). The uninfiltrated SU-8 film exhibited a large (∼34%) increase in resistance during UV exposure, which we expect was caused by the UV-induced decomposition of the charged $(Ar)_3S^+SbF_6^-$ PAG. The degree of UV-induced resistance increase was reduced with more ZnO-infiltration synthesis cycles, which is in line with the notion that ZnO accumulation near the film surface acted as a UV-blocking layer, passivating the underlying SU-8. From 0 to 4 ZnO-infiltration cycles, the percent change in resistance decreased by a factor of ∼100, agreeing with the results in Figure S9, which show that four cycles provided enough ZnO for the creation of conductive pathways near the film surface. Meanwhile, the control ALD ZnO films displayed negligible change in resistance upon UV exposure (Figure 6b), most likely due to nominal small thickness (i.e., <1.6 nm assuming 1 Å/cycle deposition rate for normal ALD), which limited UV absorption.

4. CONCLUSIONS

Infiltration synthesis of ZnO in SU-8 thin films generated organic−inorganic hybrid nanocomposites displaying UV-enhanced gravimetric sensing behavior. UV irradiation leads to reversible photowetting and improved sensing properties of SU-8 polymer alone. The gravimetric response of ZnO-infiltrated hybrid SU-8 films was ∼20 times higher than that of the uninfiltrated control SU-8 film during exposure to water vapor and oxygen. The response was enhanced by an additional 50−500% under UV irradiation. We proposed that in case of composite film, the increased adsorption sites were provided by the infiltrated ZnO nanoparticles in SU-8 matrix, in conjunction with the reversible UV-induced decomposition of the PAG in SU-8 into hydrophilic fluoroantimonic acid leading to increase in the adsorption affinity toward oxygen and water (corroborated with DFT calculations), as the primary mechanisms that enabled high gravimetric vapor/gas sensing performance of ZnO-infiltrated SU-8 film. We further demonstrated that under illumination SU-8−ZnO composites can (without additional modifications) differentiate between water vapor and oxygen gas with over 96% accuracy. The results also not only represent a new application of the inorganic−organic hybrid nanocomposite generated by infiltration synthesis as a highly sensitive gravimetric vapor/gas sensing medium, but also shed light on the possibility of using reversible UV-induced chemical changes to tune hydrophilicity for modifying selectivity and sensitivity of thin films for sensing applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.8b08072.

Irradiance of the UV lamp; AFM images, gravimetric sensor response to H_2O and O_2 in dark and under UV; water contact angle; contact potential difference images Raman and photoluminescence spectra and their analysis (PDF)

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Notes

The authors declare no competing financial interest.

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